



Decoding the Bottle: What Defines a High-Quality Wine?

DALAS Course Project

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1 Introduction

Wine, a well-known alcoholic beverage made from fermented grape juice, comes in many varieties - red, white, rosé, and more. Its quality depends on a variety of factors, including the characteristics of the production region, the grape composition, and the levels of alcohol and sugar.

As of 2023, 36 countries are recognized for wine production¹. However, with such a diverse market – thousands of bottles from different regions, made from various grape varieties, and priced anywhere from a few dollars to several thousand – how can a consumer confidently choose a bottle without regretting their choice, especially if they are not familiar with wine or lack reliable guidance from the seller?

2 Problem statement

The goal of this project is to estimate wine quality as reflected by public ratings. Previous studies, such as those using Kaggle datasets, have focused on predicting quality based on a wine’s chemical composition. However, this information is typically unavailable to the average consumer, making it difficult to judge whether a wine is worth its price.

We also aim to explore the long-standing question: does a higher price truly correspond to higher quality? In other words, do expensive wines truly taste better according to public ratings? Beyond price, we seek to identify the factors that most strongly contribute to a wine being perceived as high quality, providing insight for consumers who are less familiar with wine.

3 Data collection

To address our problem statement, our dataset will be constituted of wines’ characteristics. At first, we considered combining a dataset from Kaggle with scrapped data.

3.1 Kaggle dataset consideration

The [Kaggle dataset](#) we considered provides information on the chemical composition of around 5,000 wines. However, the dataset only covered wines from northern Portugal. Moreover, the chemical description of wines is not an information a user can easily find, nor were we able to find any website providing such in-depth composition. Given these limitations and the dataset’s restricted scope, we will rely solely on scraped data.

3.2 Web scrapping

Even though, we tried to diversify the country of production as much as possible, some countries are still underrepresented in our dataset (Russia, Brazil), and some countries are not present at all (Argentina, Turkey, China). One difficulty that came with scrapping data was to find websites that provided sufficient information about the wines. After some research, we decided to scrap our data from 2 websites: [Vivino](#) and [SimpleWine](#). The first website features wines from a lot countries in Europe, in America and very few in Africa. We didn’t find a website featuring wines in Asia that contained enough information: either tasting notes were missing or ratings were unavailable. Usually, websites provided one or the other but not both, and since we didn’t have a lot of variables to work with, we decided to that it was important to include it.

3.3 Difficulties and solution

During the web scrapping part, we encountered several difficulties.

- For the *Vivino* website, we initially considered scraping data directly from the HTML pages. However, navigating through the pages proved difficult due to occasional CAPTCHA challenges. To overcome this issue, we decided to interact with the API instead.

¹Data from this [website](#)

- As the *Vivino* API was not publicly available, we had to analyze the HTTP requests to be able to retrieve the API endpoint to collect the wines' information. Multiple different API requests were required to gather all the details for each wine.
- To minimize price discrepancies (for example, due to taxes), we tried whenever possible to scrape data by production region, since the website allows selecting the shipping country (e.g. for French wines, we set the shipping country France). We observed that the availability of certain wines varied depending on the country of origin – for instance, the list of available wines differed between France and the USA. To address this, we used VPNs to scrape data according to both shipping region and production region.
- Concerning *SimpleWine*, it was impossible to interrogate its API because of internal protection (public-private key). Since the the number of Russian wines that could possibly be scrapped didn't justify the time spent trying to bypass that protection, we directly interrogated HTML and the rendered API response `application\json` using `playwright` and `scrapy`.

3.4 Variables

Overall, we collected ~45 000 wines. Most of the data come from *Vivino*, with ~500 wines coming from *SimpleWine*. The variables available in the *Vivino* and *SimpleWine* databases are the same.

For wines from *Vivino*, the dataset is split into 2 parts: parent wines (wines without a vintage) and their children (wines with a vintage). Each parent wine can have multiple children and provides baseline characteristics shared across its vintages.

Variables present for both parent and child wines:

- | | |
|-------------------------|--|
| • Wine name | • Winery |
| • Type (color) | • Alcohol percentage, also called ABV ² |
| • Country of production | • Grapes |
| • Region of production | • Food pairings |

Variables present only for child wines:

- | | |
|--|---|
| • Vintage (year of harvesting) | • Taste (0 – 5): intensity, sweetness, acidity, tannin, fizziness |
| • Alcohol percentage, ABV ($\neq$ parent's alcohol) | • Drinking window (years) |
| • Grapes ($\neq$ parent's grapes) | • Ratings distribution (number of 1–5 stars) |
| • Volume of a bottle | • Professional rating |
| • Price of a bottle | |
| • Currency of the price | |

4 EDA and Transformation

In this section, we will do exploratory analysis to better understand the data, identify potential biases, and see what preprocessing should be done to prepare it for training predictive models. Some data transformation will also be done here to make plotting easier and wines more comparable, e.g., converting prices to a common currency or merging grapes.

Percentages shown in the histograms are always computed over the non-missing values of the corresponding variable.

²Alcohol by Volume

4.1 Missingness rate

The first thing to look at is the missingness rate for each variable. Table 1 lists all variables, their type, and the percentage of rows in which the value is missing. Some variables have a high missing rate – above 50%. However, when examining the relationships between variables in more detail, we were able to conclude that not all missing values are MCAR³, as discussed in Section 5.

Variable	Type	Missing (%)
Wine name	String	0.00
Wine type	String	0.00
Year	Integer	7.74
Country	String	0.00
Region name	String	0.01
Winery name	String	0.00
Volume	Integer	0.49
Price	Float	0.49
Currency	String	0.49
Parent grapes	List	4.05
Grapes	List	66.04
Intensity	Float	4.98
Sweetness	Float	16.87
Acidity	Float	4.98
Tannin	Float	58.72
Fizziness	Float	88.10
Ratings distribution	List	0.00
Professional rating	Float	99.60
Drinking window start year	Integer	7.88
Drinking window end year	Integer	7.93
Parent alcohol percentage	Float	0.86
Alcohol percentage	Float	42.01
Food pairing	List	7.37

Table 1: Percentage of missing values per variable and variable type

4.2 Distribution of wine star ratings

For each wine, we had a distribution of ratings: the number of 1–5 star ratings given by users. To assign a rating to each wine, we take the mean of its ratings. For 95% of wines, the standard deviation lies between 0 and 0.9, small enough to consider the mean representative. Sometimes, the standard deviation can exceed 1.5 points; this affects only a minority of wines (36), which also have fewer than 10 reviews. Ratings for these wines can be highly split - only 1–2 stars and 4–5 stars. In those cases, the mean is around 3, reflecting the mixed opinions.

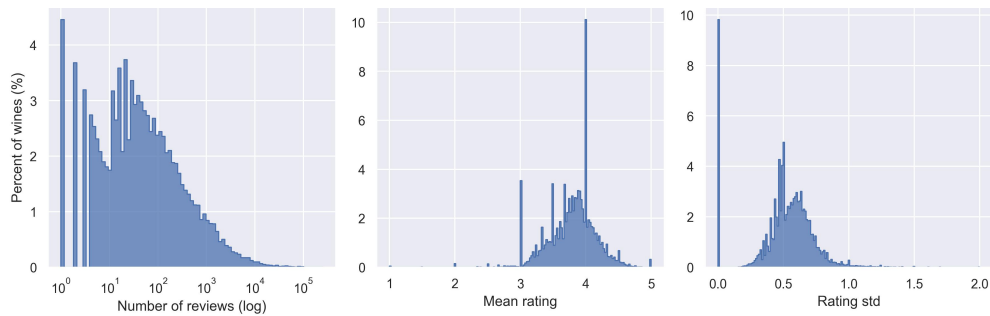


Figure 1: Rating statistics

³Missing Completely At Random

For some wines scraped from *SimpleWine*, we also have a professional rating (186 ratings in total). Figure 2 shows the distribution of professional ratings compared to user ratings. Of note, professionals tend to rate wines more harshly than users. The average difference is 0.6 points, but it can reach up to 2 points. Unfortunately, professional ratings constitute barely 1% of our database, so they cannot be used for training predictive models, especially since user and professional ratings behave very differently.

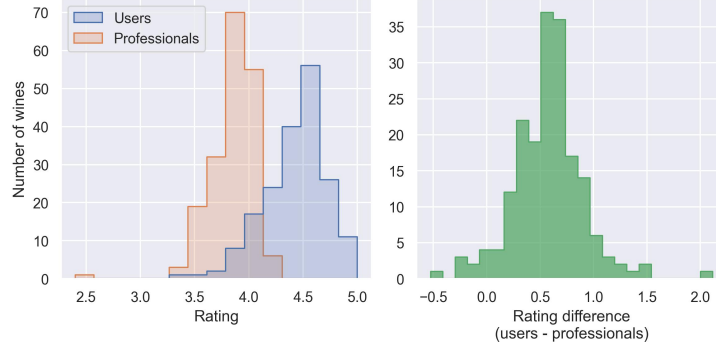


Figure 2: Users / professionals rating distributions

4.3 Price per bottle

Since we mostly scrapped data by country of production to avoid taxes (and therefore avoid measurement bias) and artificially inflated prices as a consequence, wines are priced in different currencies, as shows the figure 3a.

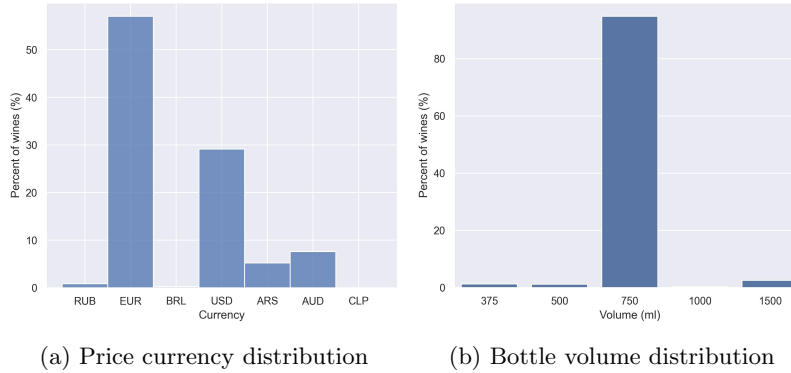


Figure 3: Currency and bottle volume distributions

To make wine prices comparable, we converted all currencies to USD, which is the standard in international trade and already accounts for 30% of our database. The conversion was done using exchange rates from the central banks of each country as of October 20, 2024. Afterward, all prices were standardized to 750ml bottles. **From this point on, all graphics use these transformed prices.**

4.4 Numerical variables

In this subsection, we analyze several numerical variables, excluding the taste variables, whose distributions are closer to categorical than truly numerical. As a result, no meaningful distributions or correlations can be expected for these variables. Figure 4 presents the joint distributions of the numerical variables along with their marginal distributions.

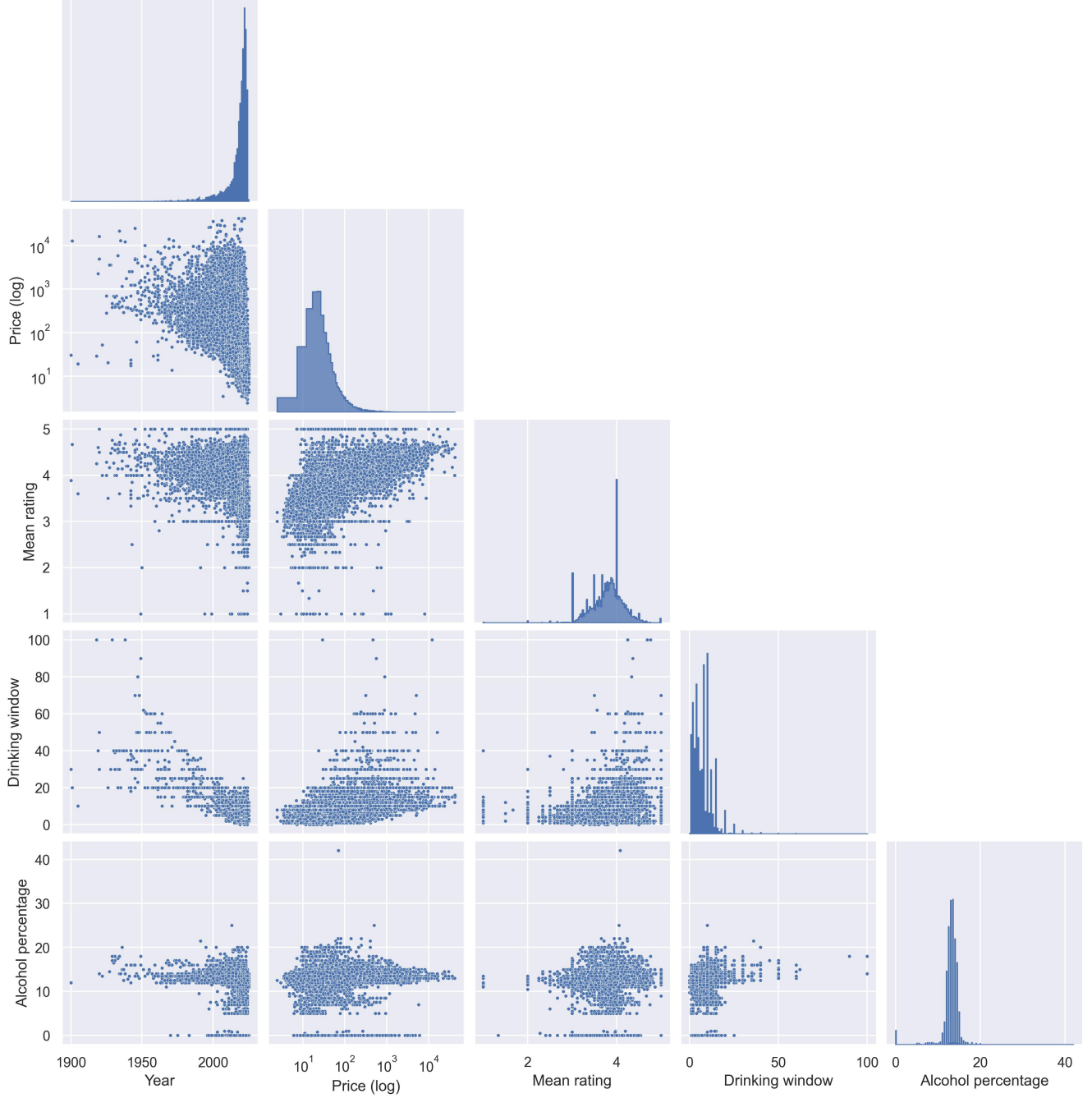


Figure 4: Pairwise and marginal distributions of the numerical variables.

4.4.1 Prices

Figure 5 shows a zoomed-in view of the price distribution already presented in Figure 4. Looking at the joint distribution of price and rating, we observe a relationship that roughly follows a logarithmic shape, the Spearman correlation coefficient of 0.72 suggests a clear positive monotonic relation between rating and price. In practice, this means that ratings tend to increase as the price increases by orders of magnitude.

However, after some research on cognitive biases [3], we suspect a presence of a prejudice bias: users tend to rate expensive wines less harshly than cheaper ones.

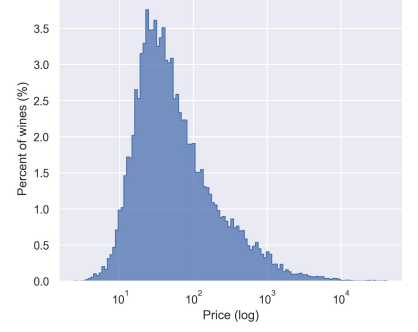


Figure 5: Standardized price distribution

4.4.2 Vintage

The distribution of vintages in our dataset is quite unbalanced. The oldest wine dates back to 1900, while the most recent one is from 2025, and 95% of the wines were produced between 1990 and 2024. Looking at the joint distribution of vintage and price, we observe that older wines generally tend to be more expensive. As a consequence of the previously identified relationship between price and rating, these wines also tend to receive higher ratings.

4.4.3 Alcohol

Since each wine in our dataset is associated with a parent wine with its own baseline alcohol percentage, we compare the alcohol distribution of vintages with that of their parent wines.

The figure 6a illustrates the distribution of alcohol percentage for vintages and for parent wines. Based on these distributions, we identified **2 outliers**: wines with an alcohol percentage above 30%, which are in fact distillates. These will be removed from the dataset before training the models. Some wines have an alcohol percentage of 0. After checking several of the affected wines on different websites, we concluded that an alcohol percentage of 0 corresponds to a missing value.

The figure 6b illustrates the difference between the alcohol percentage of a vintage and its parent wine, after removing 0 values and the 2 distillates. Overall, 46% of vintages have the same alcohol percentage as their parent, while the rest show some fluctuations, with differences ranging from -13 to +10 degrees. In theory, these fluctuations in alcohol percentage can be attributed to differences in sugar levels, either due to winemakers' practices or to changes in grape sweetness caused by seasonality [2]. However, such large discrepancies may also result from erroneous entries in the parent wine.

It is unfortunate that the websites from which we scraped the data did not provide information about sugar levels, which would have allowed us to verify this correlation. The taste variable `sweetness` is almost discrete and not descriptive enough: for example, a sweetness score of 5 can correspond to very different actual sugar levels, making it unsuitable for plotting a meaningful correlation. After looking at the joint distribution (Figure 4) with other numerical variables, alcohol doesn't seem to be correlated with either of them.

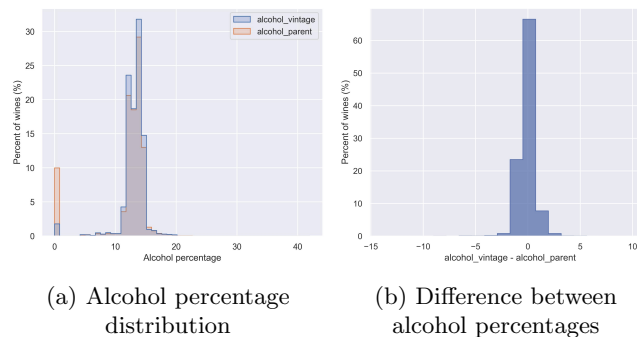


Figure 6: Alcohol distributions

4.4.4 Drinking window

On the *SimpleWine* website, the drinking window is given in years, e.g., 4 years. On the *Vivino* website, however, the drinking window is specified as a year range, for example from 2022 to 2025, which would imply having 2 inseparable variables. For this reason, we decided to convert the *Vivino* drinking window into a difference in years: for instance, from 2022 to 2025 becomes 3 years.

One would expect that expensive wines have a higher drinking window, meaning they can age longer. Distributions related to drinking window shown on Figure 4 provides some evidence for this. Some very budget wines, costing less than 10 USD, have a drinking window under 20 years. Also, the lower threshold of the drinking window gradually rises with price, suggesting that high-end wines are indeed expected to age for some time before reaching their peak.

4.5 Place of production

In our dataset, the place of wine production is described by three variables, from broadest to smallest: country, region, and winery. Figure 7a shows how many wines come from each country, while Figure 7b shows the number of regions per country. Some countries naturally produce and sell more wines, so countries like France, Italy, and the USA naturally have more wines, while countries like Moldova and Slovenia have fewer. However, countries such as Russia and Brazil are less represented than they should be.

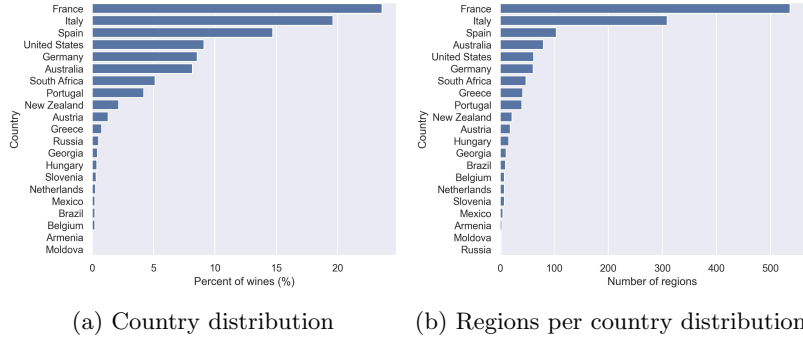


Figure 7: Place of production distributions

Each country has several production regions, and each region contains multiple wineries. As we move to more detailed levels, sparsity becomes an issue: some regions and wineries naturally have very few wines. Figure 8 shows the distribution of wines per region (subfigure 8a) and per winery (subfigure 8b), with binning applied for better visualization.

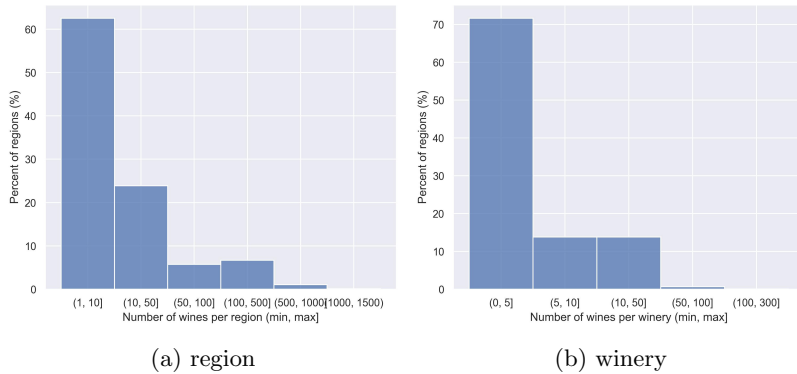


Figure 8: Distribution of number of wines per...

These figures make the sparsity obvious : more than 60% of regions have fewer than 10 wines, and more than 70% of wineries have fewer than 5 wines, even though the dataset includes 1 395 distinct regions and 8 018 distinct wineries. This sparsity will need to be taken into account when training our prediction models.

4.5.1 Winery

Although the vast majority of wineries are underrepresented, we can still create some interesting visualizations of the prices of wines from the same winery. The goal is to see whether some wineries produce mostly high-end wines while others focus on low-end wines. Figure 9 shows the distribution of mean wine prices along with their standard deviations.

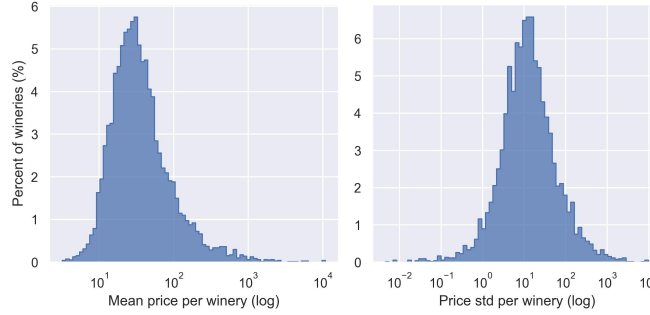


Figure 9: Price distribution per winery

Since some standard deviations are large while others are small, it's interesting to see which mean prices they correspond to. Figure 10 shows the distribution of mean prices on the x-axis and their standard deviations on the y-axis. The size and color of each point represent the number of wines for each winery.

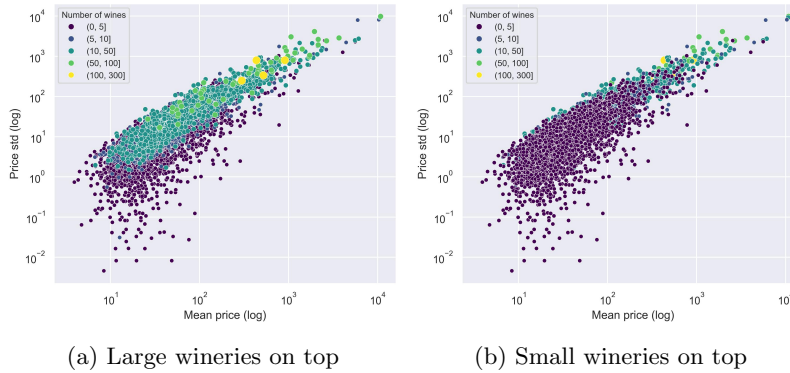


Figure 10: Wineries price means, stds

Two patterns can be observed:

- Some of the most represented (largest) wineries produce wines across a wide range of prices, from hundreds to thousands of USD.
- There are wineries – both small and large – with very small standard deviations, which indicates that some focus consistently on either high-end or budget wines.

4.6 Grapes

Before plotting, we first unified the grapes of vintages with those of their parent wines, translated grape names into English, and removed accents, ending up with 245 grapes in total. However, some grapes were very sparse: 45% appeared in fewer than 30 wines, with the rarest ones appearing in only one. Figure 11a shows the distribution of the number of wines containing each grape. To reduce sparsity, we merged grape varieties into families, based mainly on Google research, the grape database [1], the [Grapes Catalogue](#), and guidance from an LLM. These families are either popular families grown in many countries, like the Cabernet Franc, Pinot Noir, or Syrah families, or families grouping similar types, such as red hybrids, white hybrids, regional reds, and regional whites. In total, there are 61 families. Figure 11b shows the distribution of the number of wines containing each grape family.

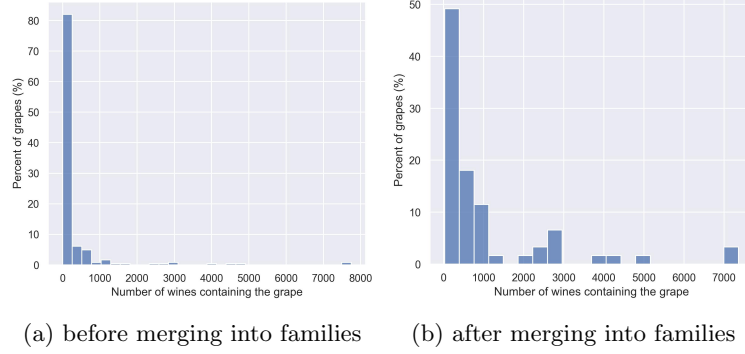


Figure 11: Number of grapes occurrences distribution

Figure 12 shows the number of wines containing each grape family. Finally, Figure 13 illustrates the co-occurrence matrix of grape families. Since some families appear much less frequently than others, the matrix is normalized by row, so each value can be interpreted as the probability that a grape from a given row is blended with another grape.

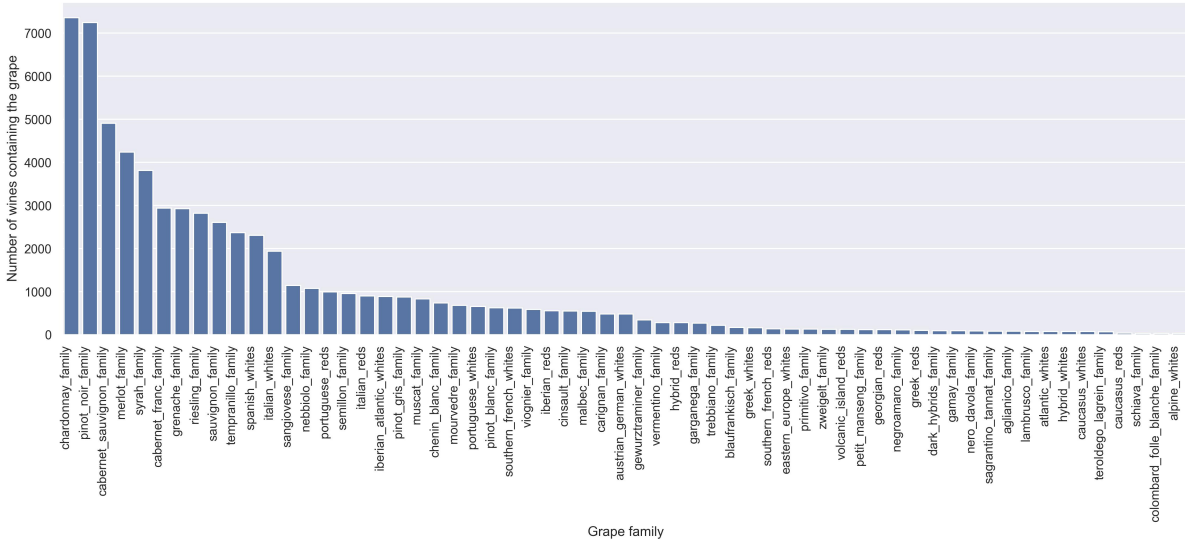


Figure 12: Distribution of the number of wines per grape family

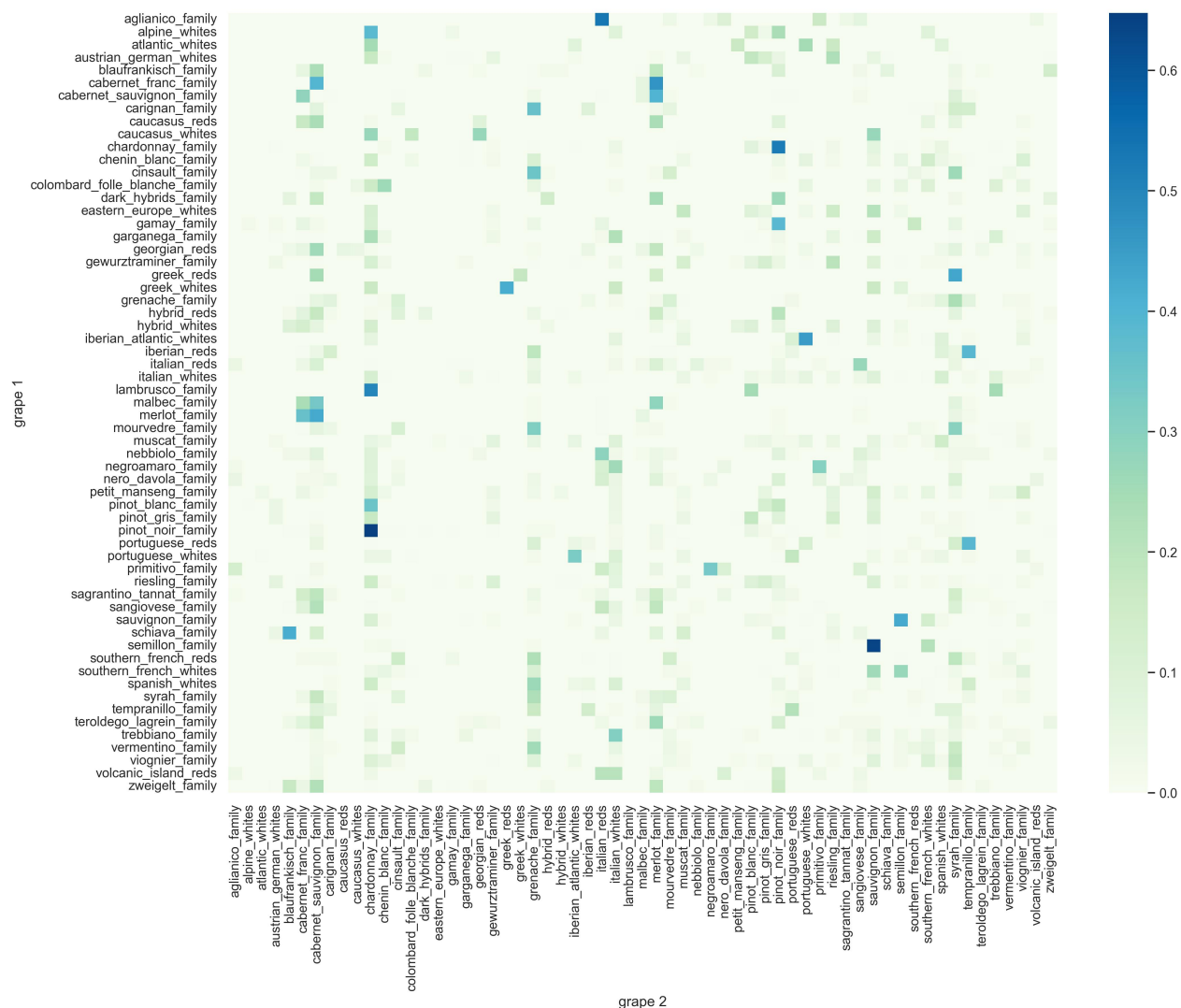


Figure 13: Normalized co-occurrence matrix of grape families

4.7 Type of wine & taste

Our dataset contains 5 variables describing taste : intensity (body), sweetness, acidity, tannin, fizziness. It is normal to expect that their distributions would vary a lot by a type of wine. The figure 14c shows how much each type of wine is represented in a dataset, and the figure 14d shows the distribution of tastes variables on a whole dataset, not taking taste into account. Each horizontal line on a violin plot indicates the observed values.

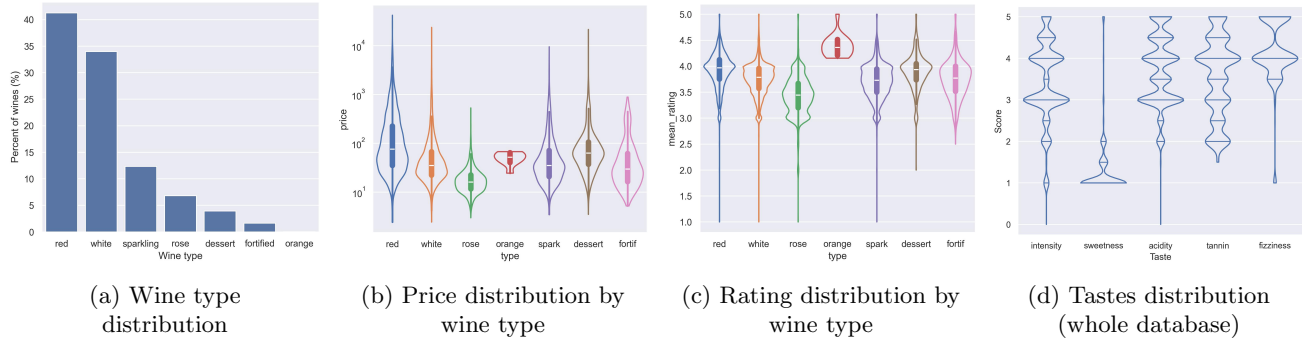


Figure 14: Wine type distributions

Figure 15 shows the distribution of the taste variables by wine type. As expected, the distributions vary by type: some are more spread out (e.g., red or fortified wines), while others are more concentrated (e.g., rose or dessert wines). Noteworthy, that the study [3] showed that taste characteristics are usually not biased by price, meaning that consumers are able to detect the correct amount of tannins, body etc. no matter the price.

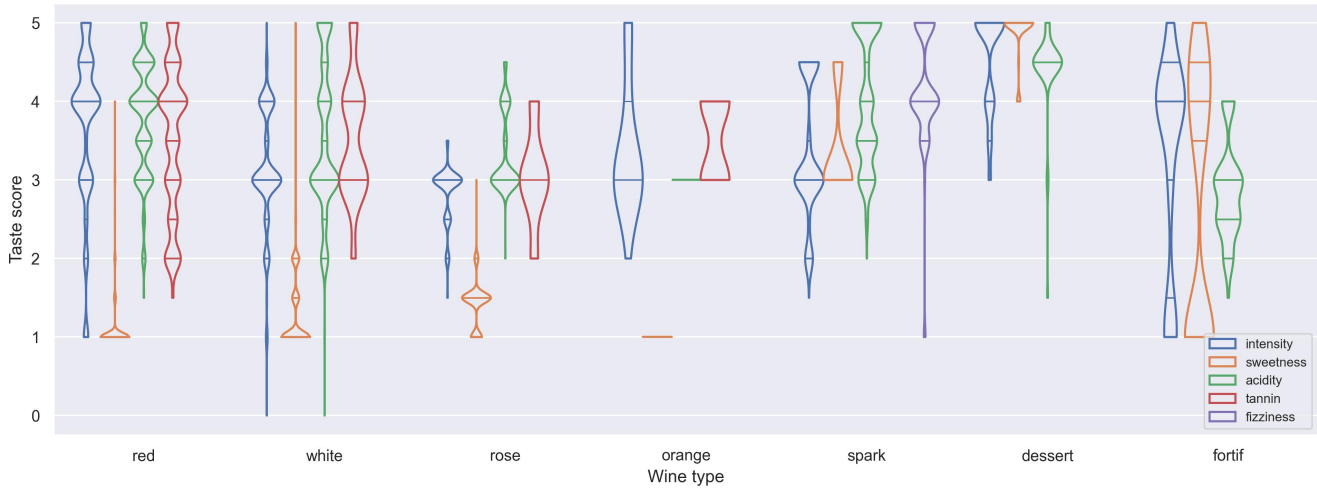


Figure 15: Tastes distribution per type

4.8 Food pairings

Figure 16 shows how well each food pairing is represented in the database. All food pairings from *SimpleWine* were merged with their English equivalents. However, the pairings *duck* and *rabbit* come entirely from *SimpleWine*, with no equivalents in the *Vivino* database. As a result, they are heavily underrepresented.

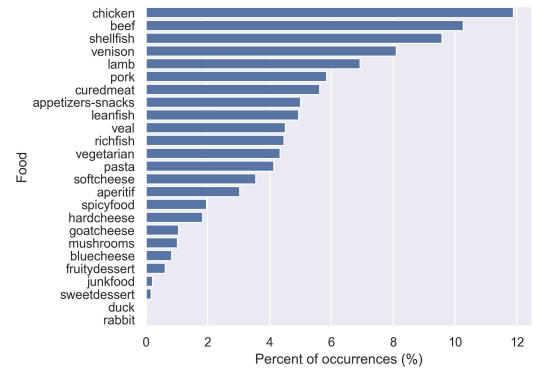


Figure 16: Food pairings distribution

It is also interesting to see which food pairings appear most often together. Figure 17b shows the co-occurrence matrix of food pairings, while Figure 17a shows the co-occurrence matrix of food pairings with wine type. Since the distribution of food pairings is far from uniform, each matrix is normalized by row. This means the values can be interpreted as the probability of a food pairing appearing together with another food pairing or with a particular wine type.

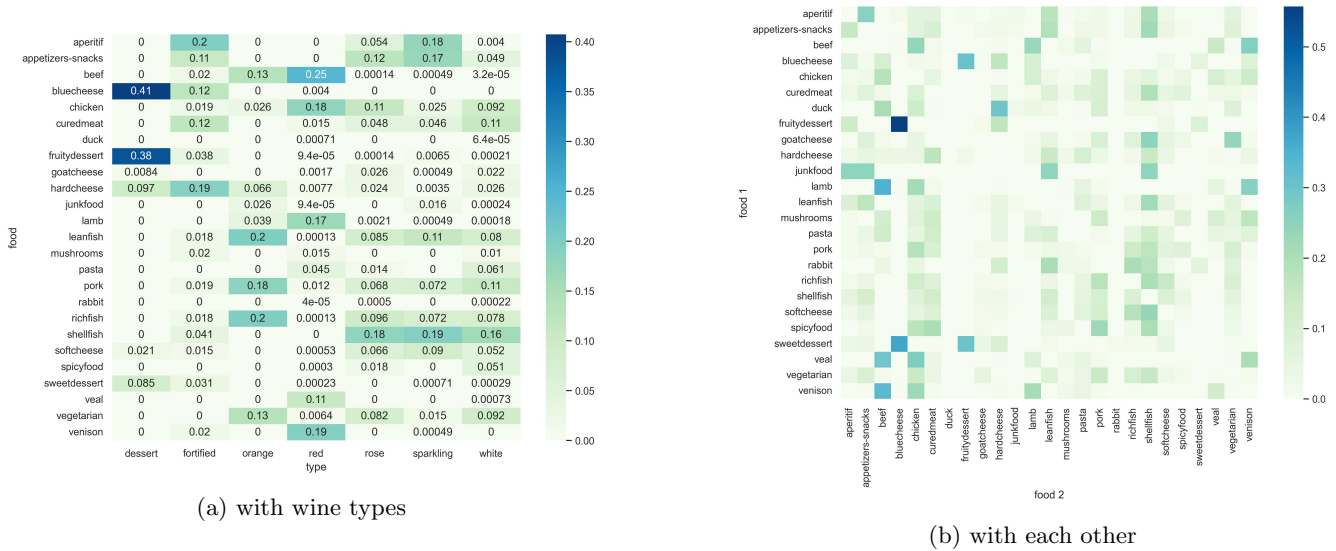


Figure 17: Normalized co-occurrence matrices of food pairings...

Some food pairings are particularly well suited to specific wine taste profiles. Figure 18 shows the mean score of each taste variable for the different food pairings. For instance, pairings such as beef, duck, lamb, rabbit, and venison are associated with wines that have higher tannin levels, whereas most other foods tend to pair with wines lower in tannins. Similarly, pairings such as fruity desserts, sweet desserts, and blue cheese are more commonly associated with sweeter wines.

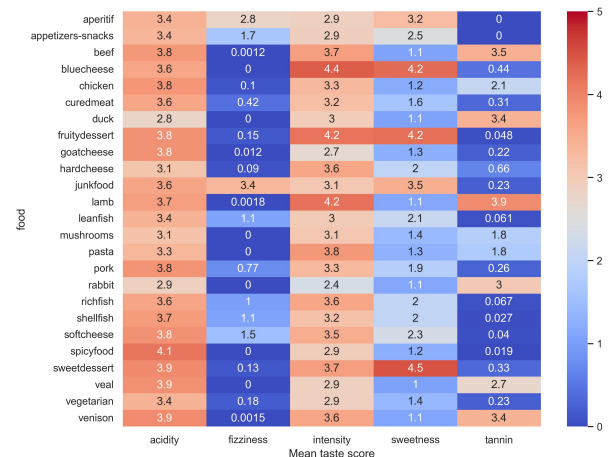


Figure 18: Mean taste score per food pairing

5 Imputation

In this section, we present the imputation applied to the alcohol percentage, drinking window, and taste variables (intensity, sweetness, acidity, tannin, and fizziness). We compare the distributions before and after imputation to ensure they remain similar, and we discuss the reasoning behind the choice of the imputation strategy.

5.1 Alcohol percentage

As noted in Section 4.4.3, an alcohol percentage equal to 0 corresponds to a missing value and will therefore be imputed.

Since alcohol percentage depends on winemakers' practices and seasonality, it may vary across vintages. Our approach is to use the parent wine's alcohol percentage as a baseline and combine it with the median alcohol percentage of the other vintages for which this information is available. As the number of known vintages increases, more weight is given to their alcohol percentages. In practice, this corresponds to taking a weighted average between the parent wine's alcohol percentage and the median alcohol percentage of its vintages.

$$a_{\text{inferred}} = \frac{(1 - w) \cdot a_{\text{median of vintages}} + w \cdot a_{\text{parent}}}{1}$$

with $w = \frac{1}{1 + |\text{vintages with known alcohol}|}$

a : alcohol percentage
 $|\cdot|$: the cardinality of a set

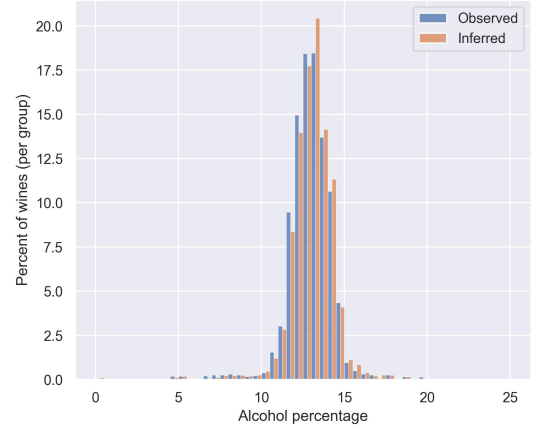


Figure 19: Distribution of observed/imputed alcohol

Figure 19 shows the distribution of observed and imputed alcohol percentage values. The distributions are similar across groups, indicating that the imputation performs well.

5.2 Drinking window

Some of the missing values are MCAR, while others come from the fact that the end year of the drinking window is not specified on the *Vivino* website. Since no wine can age indefinitely, we fill these missing values using the median. Figure 20 shows the distribution of the drinking window before and after imputation.

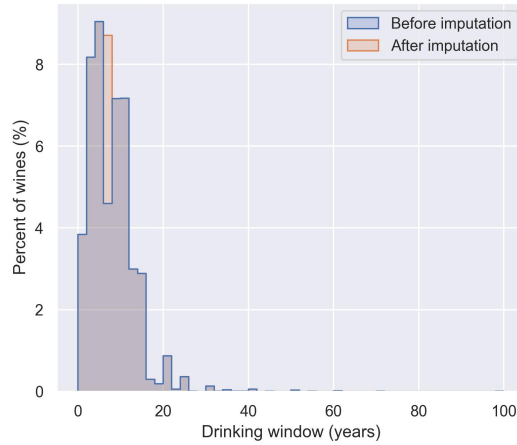


Figure 20: Distribution of observed/imputed drinking window

5.3 Taste

In this section, imputation is performed before merging grapes into families, in order to preserve as much precision as possible during the imputation process

The idea behind taste imputation comes from how taste is defined on *Vivino*: taste scores are assigned at the blend level, where a blend is defined by the country, wine type, and the principal component, grape composition. As a result, wines belonging to the same blend share exactly the same taste scores.

In Section 4.1, we observed that some taste variables have a very high missingness rate. However, this missingness is not entirely MCAR. In fact, some values are MAR, as they depend on the wine type. Figure 21 shows 2 matrices: missingness by country (left) and missingness by wine type (right), where 1 means all values are missing and 0 means all values are present.

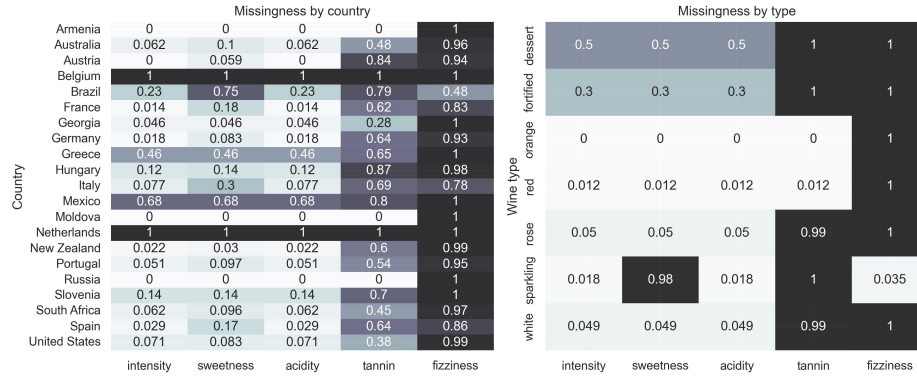


Figure 21: Taste variables missingness matrices

The missingness matrix by country of production (left) shows that, for some countries, such as the Netherlands and Belgium, taste information is simply not provided. This means that for certain regional grapes, taste data is entirely unavailable.

From the missingness matrix by wine type (right), we can make several observations:

1. Fizziness is only defined for sparkling wines.
2. Tannin is only defined for orange and red wines. The very high missingness rate (0.99) for rose and white wines comes from the fact that only the *SimpleWine* database provides non-zero values. On the *Vivino* website, these characteristics are not defined.
3. Sweetness is not defined for sparkling wines, except for a small number of them (about 2%).

These observations allow us to safely put 0s for these wine types and these taste variables. For other taste variables and wine types we will work on grape level. The idea is to find the most similar blend using the Jaccard distance, excluding cases where the distance equals 1, which indicates no overlap. Since, on *Vivino*, a blend is also defined by wine type and country, we first investigate whether it is necessary to include these variables in the grouping during imputation. To do so, we explore the mean taste scores per grape, first grouping by wine type and then by country.

Most of the imputations concern variables intensity, sweetness, and acidity for dessert and fortified wines. For this reason, the following analyses mainly focus on these variables and wine types. Figure 22 shows an excerpt from the table of mean taste scores per grape, grouped only by wine type. Taste profiles differ significantly across types, with differences reaching up to 4 points, indicating that wine type plays a major role in taste definition.

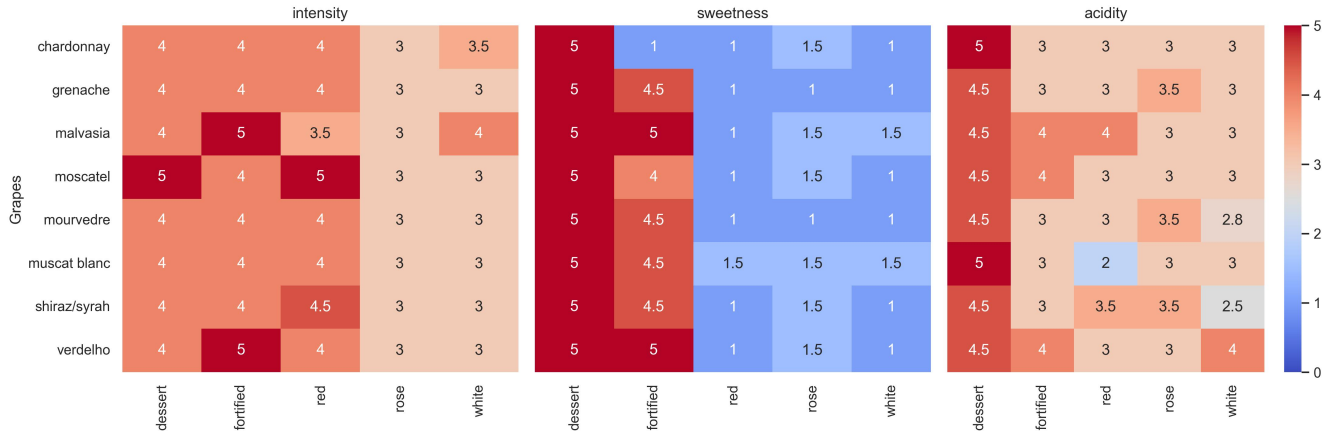


Figure 22: Mean taste score per type per grape (excerpt)

Figure 23 presents an excerpt from the table of mean acidity scores for fortified wines, this time grouped by country. For the same grape, taste scores can differ by as much as 1.5 points, for example for the grape Souzao between Portugal and South Africa. However, when grouping simultaneously by type and country, less than 5% of missing values can be imputed, due to the lack of sufficient data at this level of granularity.

As a result, although country influences taste, the data sparsity makes it impractical to rely on both type and country during imputation. Given that blends on *Vivino* are defined per type and country, we therefore prioritize grouping by wine type only and accept the resulting approximation error. Figure 24 shows the missingness rate matrix after imputation, illustrating the decrease of missingness rates that we were able to achieve using this strategy.

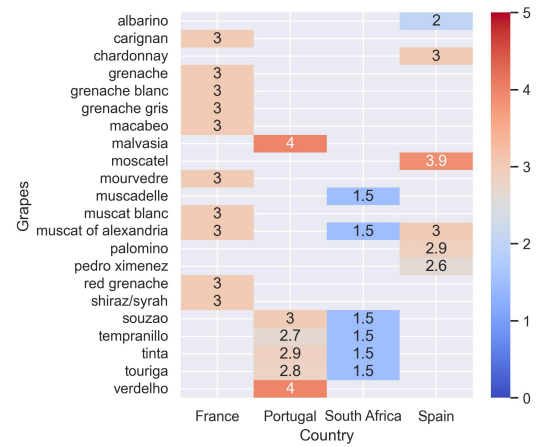


Figure 23: Mean acidity of fortified wines per grape per country

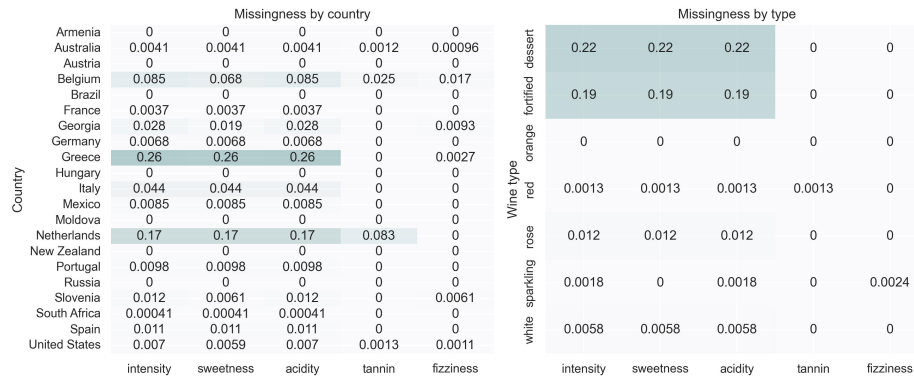


Figure 24: Missingness matrices after imputation

6 Model training

In this section, we focus on our main objective: predicting the ratings users would give to a wine based on its characteristics. We split the dataset into training and test sets and prepare the variables for predictive modeling. Since the dataset contains many categorical variables, we explore tree-based models capable of handling both numerical and categorical inputs, including **Random Forest**, **XGBoost**, and **CatBoost**. We start with the regression task and then move on to classification, where we aim to mitigate ratings' prejudice bias related to price.

6.1 Data preparation

Based on an initial assessment of the variables in our dataset and after some pre-processing, we excluded the following columns from the list:

- Wine name : each wine has a unique name, so the number of distinct names is nearly equal to the total number of wines
- Volume of a bottle : the price is associated to a volume, we converted all price to match the same volume (750 ml)
- Currency of the price: all currency were converted into USD
- Professional rating : less than 1% of the wines in the dataset has this value
- Number or reviews: isn't a wine's inherent characteristic

6.1.1 Features

We end up with the following features for our models:

- **Categorical variables:** type, country, region name, winery name, food, grapes
- **Numerical variables:** year, intensity, sweetness, acidity, tanning, fizziness, alcohol, drinking window, price
- **Target variable:** Mean ratings

For Random Forest model, string-type categorical variables have to be encoded as integers. We look at the number of distinct values of each of these variables to determine the appropriate encoding method. We choose between one-hot and target encoding, as neither assumes an ordinal relationship. For variables with a small number of distinct values, one-hot encoding can be used to expand them into multiple columns.

Variable	Number of distinct values	Encoding
Type	7	One-Hot
Country	21	One-Hot
Region	1429	Target
Winery	8543	Target
Food	25	MultiLabel Binarizer
Grapes	61	MultiLabel Binarizer

Table 2: Encoding of the different categorical variables

The variables food and grapes are multi-categorical, and has to be encoded for all of our models. To encode them, we use a MultiLabel Binarizer, which is similar to one-hot encoding but allows multiple categories per observation.

6.2 Train and test split

To maintain the same distribution in our training and test datasets, we split the data stratified by country. As shown in Figure 25, this stratification not only preserves the country distribution but also maintains the distribution of wine types. We divide our dataset into a training set (80%) and a test set (20%). Model validation will be performed on the training set, which will be further split into training and validation subsets following the same 80/20 split.

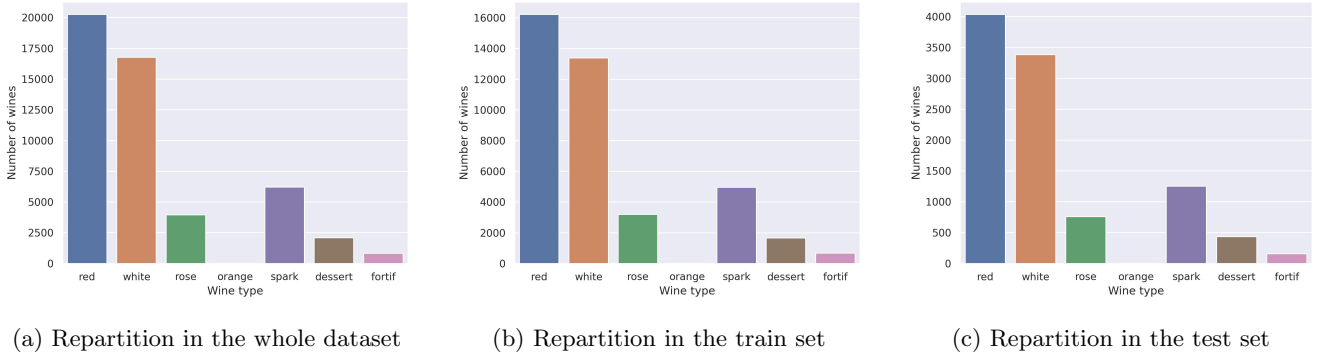


Figure 25: Repartition of wine type with stratification by country

6.3 Regression models

In this section we will evaluate our regression models using the following metrics: Mean Absolute Error (MAE) which treat all errors equivalently, Mean Squared Error (MSE) which penalizes more larger errors, and R2, which indicates the proportion of variance explained by our model.

6.3.1 Cross validation on train set

We use 5-fold cross-validation to split the data into multiple training and validation sets for the evaluation. This helps make our performance metrics more robust, reducing the risk that our results are biased by a particular train-validation split.

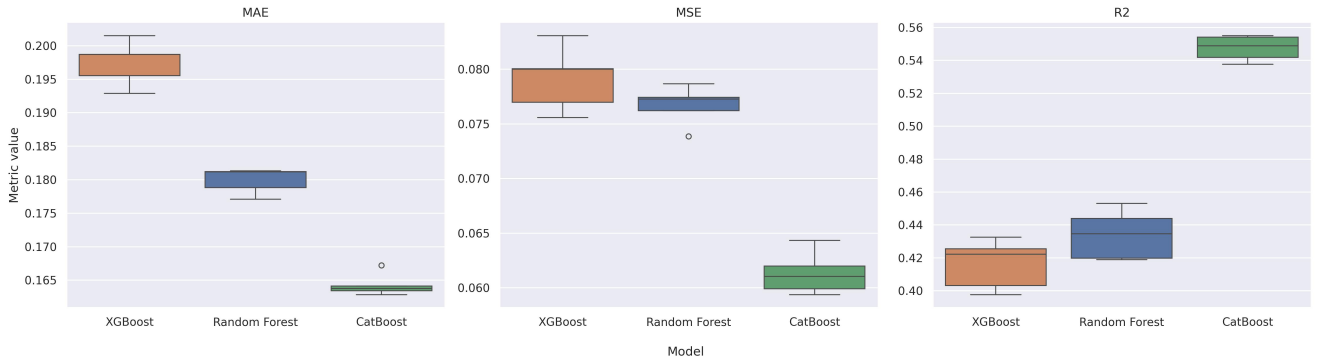


Figure 26: Model performance over 5-fold cross-validation

Figure 26 presents the performance of our three models across different metrics. For MAE and MSE, lower values indicate better performance, while for R2, higher values are preferable. Overall, the CatBoost model consistently demonstrates the best performance.

6.3.2 Performance on test set

We evaluate the best performing model on the reserved test set. With CatBoost, we obtain the following results:

	MAE	MSE	R2
Train set	0.157	0.056	0.588
Test set	0.1626	0.0593	0.552

Table 3: CatBoost performance on train and test sets

As shown in Table 5, our model performs very similarly across all metrics on the training and test sets. If the R2 and accuracy is slightly better in the train set, the very close similarity indicate that our model generalizes well and do not overfit strongly during training.

Figure 27 shows the predictions of our model for each known rating. We can notice that predictions are less accurate for low ratings (below 3) as well as for the highest rating (5), as expected from the rating distribution as seen in Figure 1, as these ratings are infrequent. For the more frequent ratings, we can see a tendency of the points around the $x = y$ line, indicating that our model captures the overall trend.

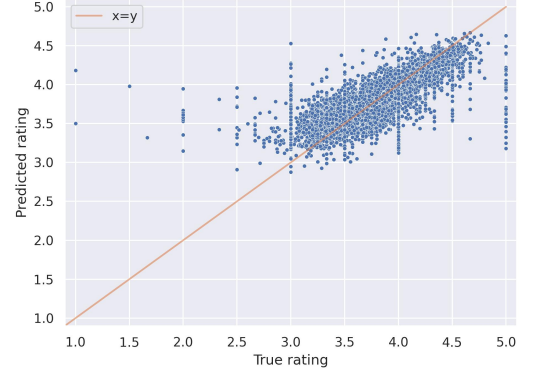


Figure 27: True versus predicted ratings on the test set using CatBoost

6.3.3 Price bias

As discussed in Section 4.4.1, we suspected a cognitive bias in ratings related to price. We decided to see how the model reacts when we remove price as a feature, meaning the model will be trained only on features, which are supposed to represent wine’s intrinsic quality. To compare predictions from models trained with and without price, we compute the residuals defined as $\text{Residuals} = \text{True rating} - \text{Predicted rating}$. Figure 28 illustrates the distribution of these residuals.



Figure 28: Residuals with and without including the price as a feature with XGBoost

Figure 28a shows that for expensive wines, the residuals tends to be above 0, meaning that the predicted ratings were lower than the true ratings. In contrast, in Figure 28b, we can notice that including the price reduces the residuals: the predicted ratings are higher and more closer to the true ratings. Moreover, as the price increase, the residuals get closer to 0, showing a higher predicted rating and showing the presence of a modeling bias related to the price. Hence, the model can’t capture users preferences without price, which may be a consequence of a cognitive bias or the fact that price acts as a proxy variable for unobserved quality markers, such as superior soil conditions or vine age.

6.4 Classification models

Our next goal was to try and deal with a cognitive bias relative to price. In order to do that, we would reframe our problem into a classification task. As we want to predict the intrinsic quality of a wine, the idea here is to predict if a wine is better or worse than the average wine in a specific price category, which is a binary classification problem.

First, we normalize wine ratings by binning wines according to their log-prices and subtracting the average rating within each bin. Price is then excluded when training the models. This adjustment highlights wines that are rated higher than others at similar prices – potential ”hidden gems” – as well as wines rated lower than their peers, which may be overpriced or not worth their cost.

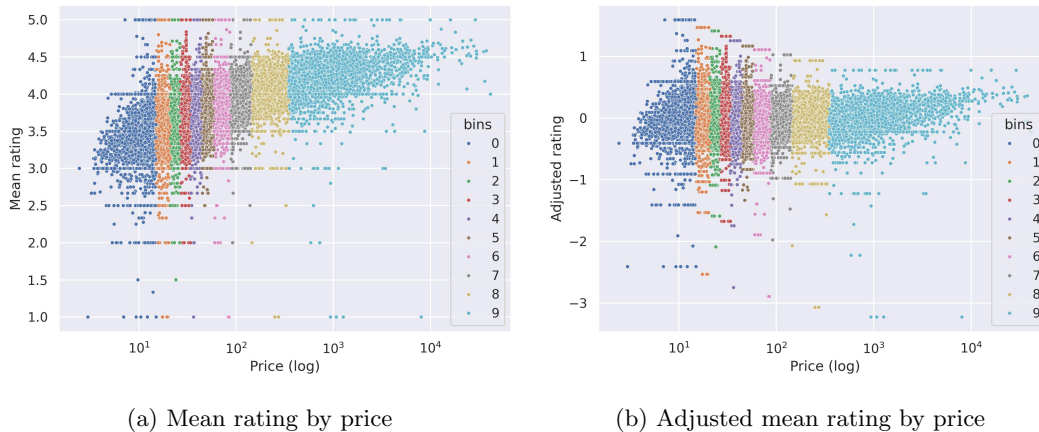


Figure 29: Mean ratings before and after removing price bias, for 10 bins

To bin wine prices, we use decile (10-quantile) binning, resulting in approximately 4 000 wines per bin, which is large enough to compute reliable mean ratings. The resulting price bins are: [0, 2, 14, 20, 26, 33, 43, 58, 86, 145, 347, 41573]. Of note, these bins roughly align with common price-based wine categories defined by [Wine Folly](#), whose book we used as a reference in this project. Their categorization is based on an analysis conducted in 2005 and adjusted for inflation in 2016. Our price ranges are generally higher, reflecting more recent inflation, and may not fully align with Wine Folly’s categories, as our dataset includes wines from multiple countries – not only the USA – and prices were converted from local currencies into USD, which may introduce additional differences.

Figure 29a shows the division of our `train` data into 10 bins, grouping wines with similar price ranges together. In Figure 29b, we adjust the ratings by subtracting to each rating, the mean rating of the price bin it belongs to, which allows for a fair comparison of wines within the same price category after removing the price bias.

6.4.1 Cross validation on train set

After adjusting the mean ratings, we remove the price from the features and run our models to see if they can correctly predict whether a wine will have a higher rating than others from the same price range. We will evaluate our models using the F1 score, which balances precision and recall, as well as accuracy. Figure 30 presents the performance of our three models across the two metrics. Overall, CatBoost achieves the best results, with Random Forest performing closely behind.

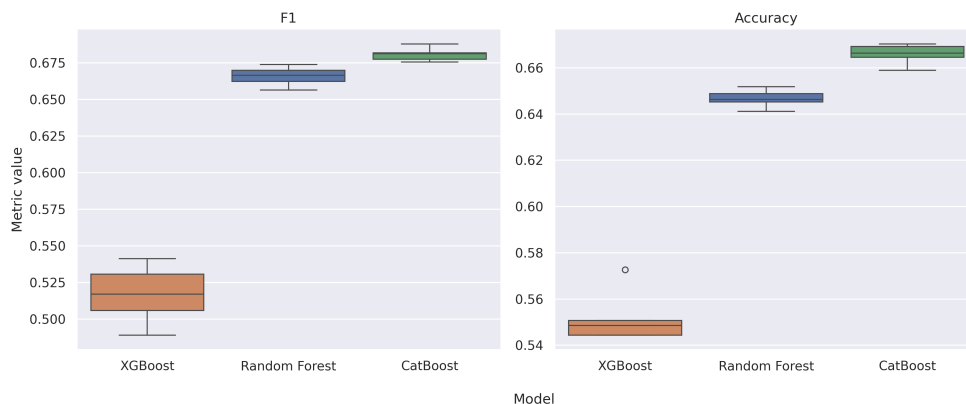


Figure 30: Model performance over 5-fold cross-validation

6.4.2 Performance on test set

We evaluate our CatBoost model on the test set, obtaining the results in Table 4.

	F1	Accuracy
Train set	0.732	0.717
Test set	0.684	0.667

Table 4: CatBoost performance on train and test sets

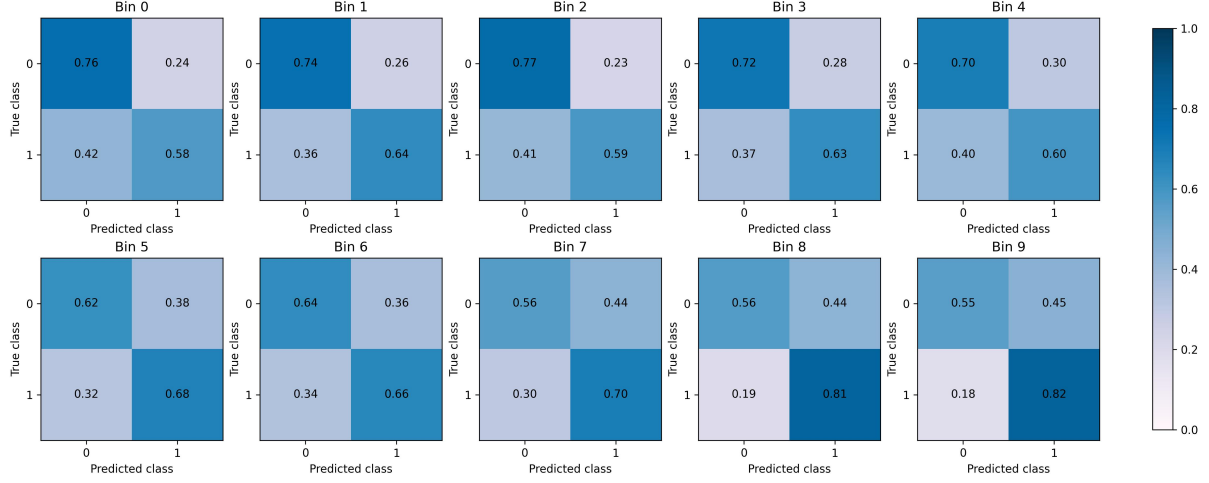


Figure 31: Confusion matrix for all bins in the test set

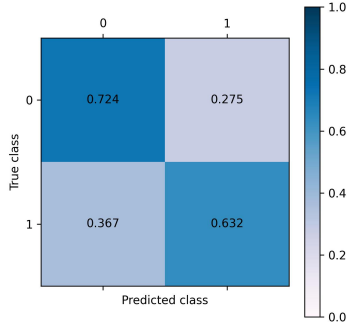


Figure 32: Confusion matrix for the test set

Figure 31 shows the confusion matrices for each price bin. In the five lowest price-range bins, the model more accurately predicts class 0, corresponding to wines with ratings below the bin mean. In contrast, in the three highest price-range bins, the model performs better at predicting class 1, representing wines with ratings above the bin average, while performance in the middle price ranges remains largely unchanged across classes.

Figure 32 presents the confusion matrix aggregated over all bins, showing that class 0 is predicted more accurately than class 1.

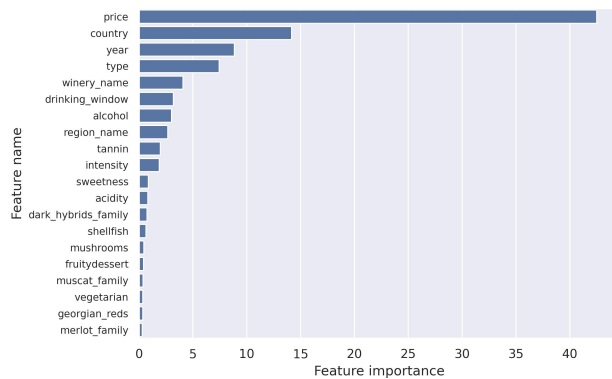
7 Model Interpretability

In this section, we aim to understand the factors driving our models' predictions for both the regression and classification tasks. For classification task, we are particular interested in identifying which features most strongly influence the models' outputs

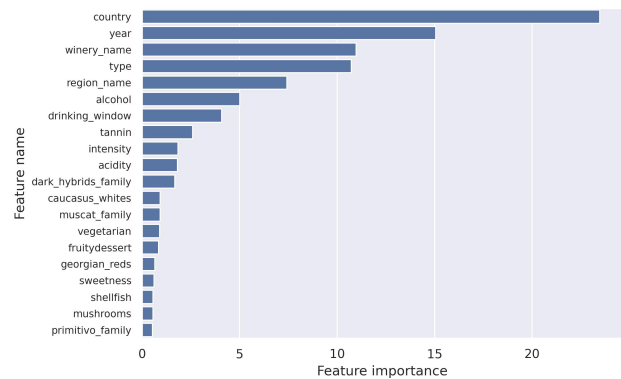
7.1 Feature importance

The CatBoost model provides feature importance scores, which allow us to assess the influence of each feature on the predictions.

7.1.1 Regression



(a) regression



(b) classification

Figure 33: Top 20 most influential features for...

Figure 33 illustrates the contribution of features to the models. As expected, the price has a strong influence on the regression predictions. Of note, in both models, the top 10 influential features are largely the same, with only slight differences in their ranking.

While feature importance shows which features the model relies on overall, it doesn't reveal in which direction it influences the prediction, or how it affects an individual prediction. For instance, we can see that the year influences a lot in the predictions, but we do not know whether an older vintage contributes to increase or decrease the prediction. For this, we will use the SHAP values that provides a more detailed view of the factors behind the model's decisions.

7.2 SHAP Values

SHAP (SHapley Additive exPlanations) values help us understand the contribution of each feature to individual predictions. Since we have both a regression and a classification task, we will examine the SHAP values for each.

7.2.1 Regression task

Figure 34 shows the SHAP values for the most influential features. We observe that the results closely align with the feature importance rankings, particularly for the top features. Each dot for a feature corresponds to a wine in the dataset and the color shows the variable's value relative to the dataset. Non numeric values, corresponding to string-type categorical variables without ordinal order, are represented by gray dots.

We observe that a higher **price** generally increases the predicted rating, further supporting the presence of a prejudice bias. Similarly, an older **vintage** (lower feature value) tends to raise the predicted rating, while a younger vintage lowers it. For **alcohol** content, higher percentages are associated with higher ratings, and lower percentages with lower ratings. Finally, **drinking window** shows mixed effects: lower values sometimes increase the predicted rating, though exceptions occur.

Figure 35 shows the distribution of SHAP values for taste attributes and wine type. For **taste-related** features, the presence of tannins tends to increase the predicted rating, while sweetness generally lowers it. Higher fizziness values also influence predictions, whereas acidity and intensity show mixed effects, with higher or lower values dominating depending on the case.

Regarding **wine type**, the predicted ratings show a mixture of increases and decreases overall. Specifically, the rosé type tends to lower predictions, whereas orange (rare value) and dessert wines tend to increase them.

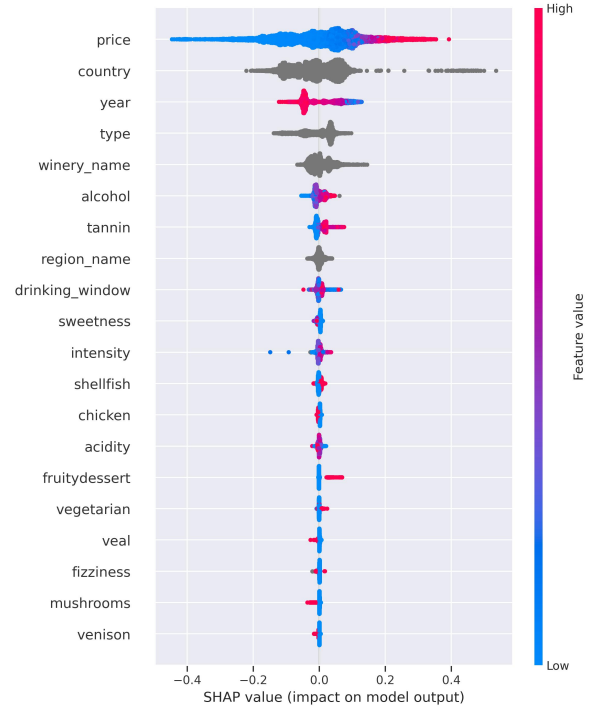


Figure 34: Top contributing features

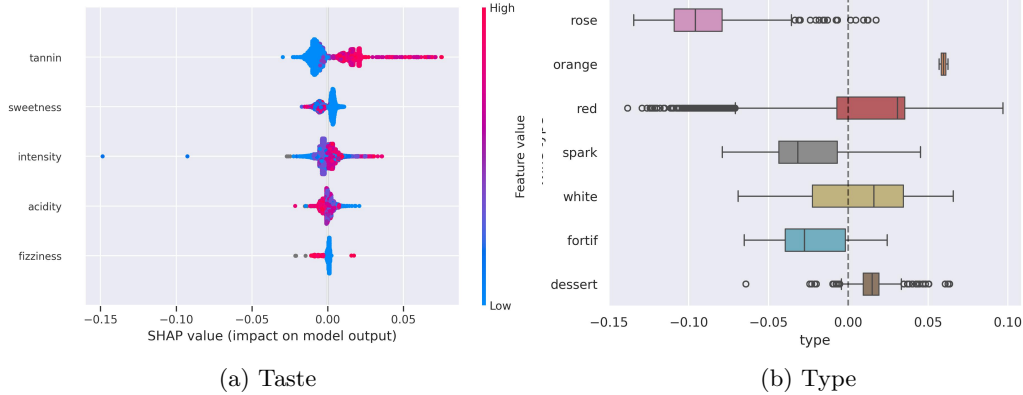


Figure 35: SHAP values for taste and wine type

Figure 36 shows the SHAP values distributions for the food and grapes variables. For **food pairings**, the absence of a suggested pairing generally has little effect on the predicted rating. Certain foods, however, do influence predictions: mushrooms tend to lower the rating, fruity desserts tend to raise it, and some pairings can have either positive or negative effects depending on the wine. A similar pattern is observed for **grapes**.

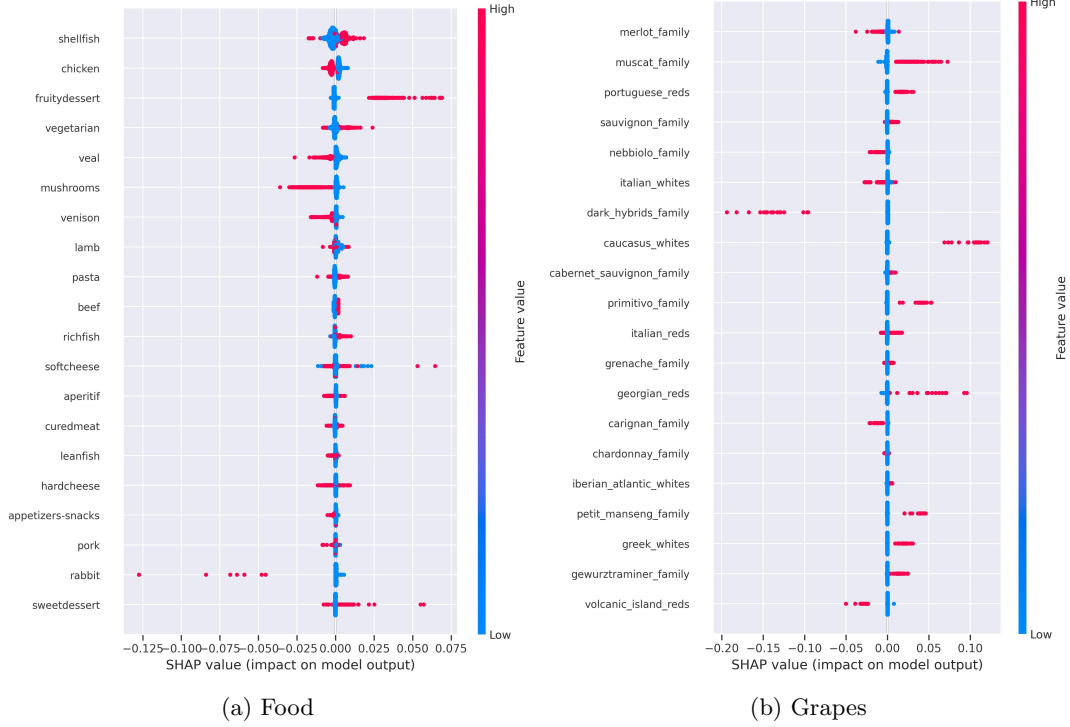


Figure 36: Most influential SHAP values for food pairing and grapes

Figure 37 shows the SHAP value distributions for the most frequent countries and regions. Some categories have high SHAP values due to very few samples – for example, Armenia has only 25 wines in the dataset. For this reason, we do not display features ranked by highest mean absolute SHAP values.

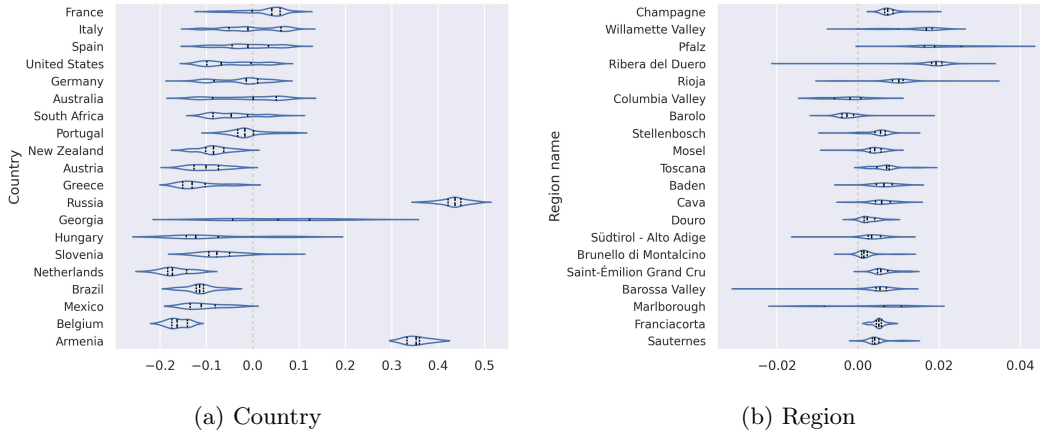


Figure 37: Most influential SHAP values for country and region

Figure 38 also presents some noteworthy dependency plots of variables by price. In subfigure 38a, we see that vintage years between 1980 and around 2018 tended to increase predictions, especially for higher prices. However, after 2018, this trend reversed, with these years contributing to a decrease in predictions. Moreover, higher prices were linked to a greater reduction in predictions.

In subfigure 38b, wines with an alcohol content around 13% show a decrease in predictions, while both lower and higher alcohol percentages lead to an increase, possibly due to their frequency. High prices increase more significantly than low prices near this value, but this effect is less pronounced at other alcohol levels.

Finally, in subfigure 38c, a longer drinking window is associated with higher prices, with the effect varying overall. However, for very long drinking windows, the prediction tends to increase predictions.

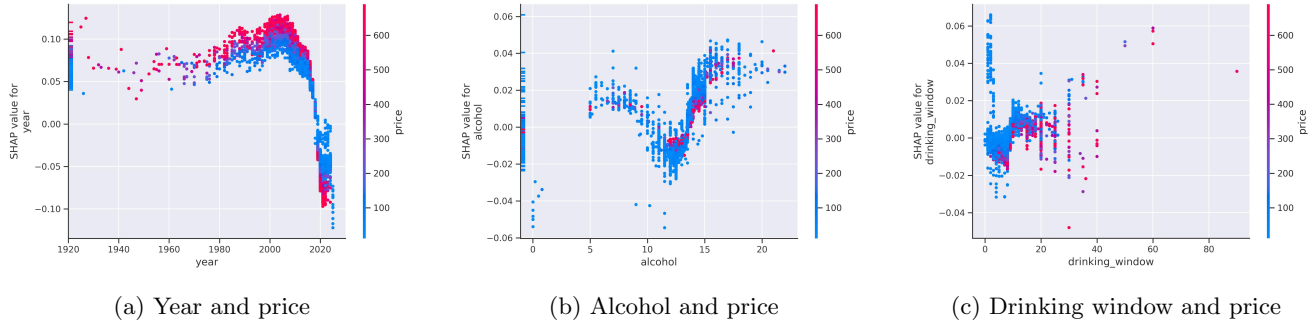


Figure 38: SHAP dependency plots for year, alcohol, and drinking window by price

7.2.2 Classification task

We realize a similar SHAP values analysis for the classification task model.

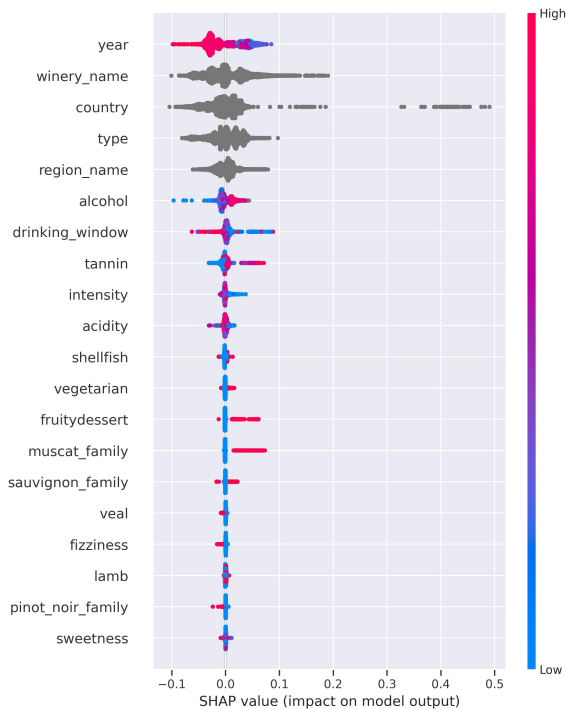


Figure 39: Top most contributing features

Figure 39 presents the SHAP values of the most important features. The categorical variables do not appear on the color scale, and the distribution of values is slightly left-skewed due to some extreme SHAP values associated with the country variable.

We observe a similar pattern as seen in the regression model for the variables vintage year and alcohol, as well as for the drinking window, although the latter seems to favor extreme values at both ends. Additionally, the most influential features are also consistently ranked at the top.

Figure 40 shows the distribution of SHAP values for taste attributes and wine type. For **taste-related** features, higher levels of tannins are associated with an increased predicted rating, while lower levels tend to decrease it. Lower intensity generally leads to higher predictions. The presence of fizziness tends to decrease the predicted rating. For acidity and sweetness, the relationship is more mixed; for acidity, high and low values appear slightly more often at the ends, but neither extreme is predominant.

Regarding **wine type**, the predicted ratings show a mixture of increases and decreases overall. We notice a similar behavior as before, notably the increase in the values for under represented types.

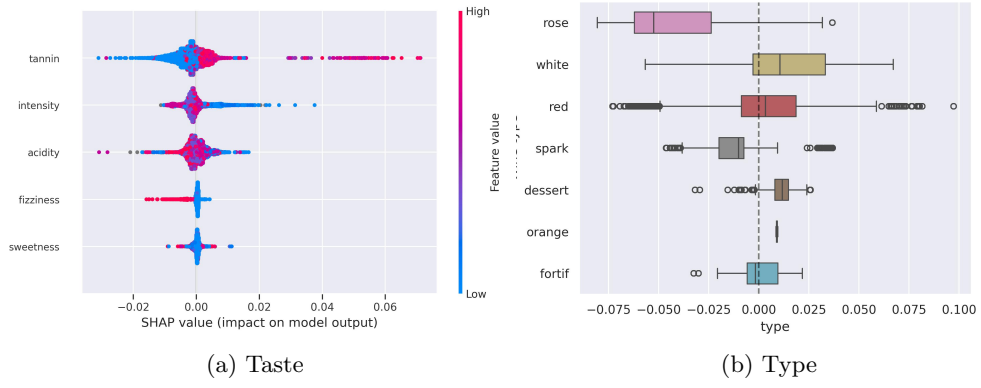


Figure 40: SHAP values for taste and wine type

Figure 41 illustrates the distribution of SHAP values for the **food pairings** and **grapes** variables, showing patterns that generally align with those observed previously. Similarly, Figure 42 presents the SHAP value distributions for the most common **countries** and **regions**, which largely follow the same trends.

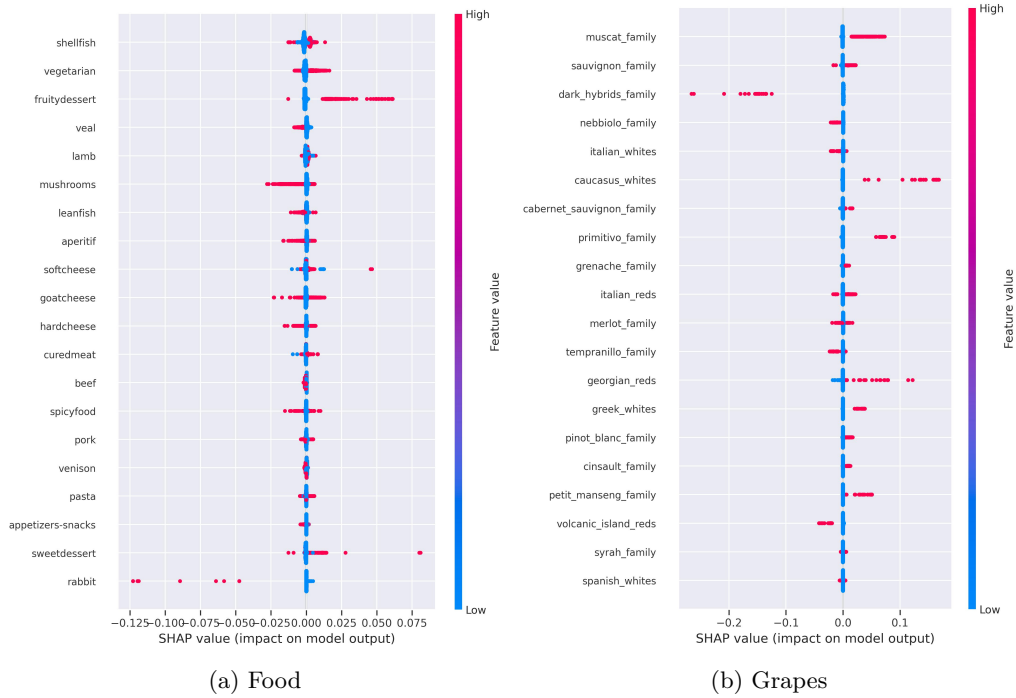


Figure 41: Most influential SHAP values for food pairing and grapes

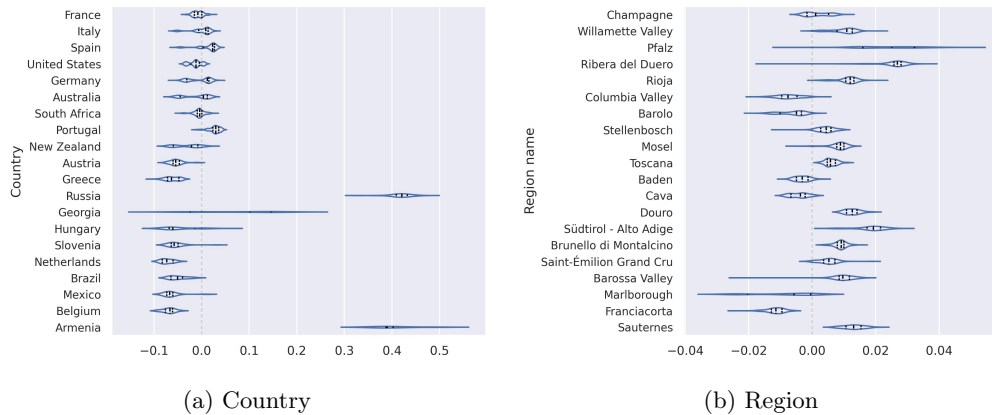


Figure 42: Most influential SHAP values for country and region

8 Conclusion

In this project, using scrapped data from *Vivino* and *SimpleWine*, we explored whether it is possible to predict wine quality based solely on user-accessible ratings. The answer is yes: it is indeed possible to understand whether a wine is worth trying given its characteristics, although with some important remarks regarding price bias, as previous studies [3] have shown that people naturally tend to rate expensive wines more favorably than budget wines, due to cognitive expectations.

One of the main questions at the start of the project was whether expensive wines truly taste better. This is difficult to answer with certainty. While expensive wines generally receive higher ratings, this may reflect a combination of cognitive bias and other factors, such as soil quality, grape-growing practices, or winemaking techniques, for which price can act as a confounder.

Accordingly, our analysis considered not only a regression task to predict ratings, but also a classification task to determine whether a wine is a "hidden gem" or overpriced compared to other wines in the same price range.

For both tasks, CatBoost outperformed XGBoost and Random Forest. The regression model had limited success: low ratings were overestimated, and high ratings underestimated, likely due to the lack of data at the extremes. To address price bias, we focused on the classification task, predicting whether a wine is better or worse than the average in its price category, achieving an F1-score of around 0.7 on the test set.

Model interpretability was enhanced through SHAP value analysis, which provided insight into the importance and impact of each feature on individual quality predictions.

Future work could focus on more NLP-based tasks using user reviews and on improving user-oriented wine recommendations. In this project, we did not scrape review data, mainly due to the challenges of handling infinite scroll, but with more time it would be feasible. Additionally, an interactive app available for end users could be developed to provide insights into why a particular wine was predicted to be better or worse than the average wine in its price category.

References

- [1] Kym Anderson and Nanda R. Aryal. *Which Winegrape Varieties are Grown Where?: a global empirical picture*. University of Adelaide Press, 2013. ISBN: 9781922064677. URL: <http://www.jstor.org/stable/10.20851/j.ctt1t304tz>.
- [2] Madeline Puckette and Justin Hammack. *Wine Folly: The Essential Guide to Wine*. 1st. New York, NY: Abrams, 2015. ISBN: 978-1419724706.
- [3] Charles Spence. “Cognitive influence on the evaluation of wine: The impact and assessment of price”. In: *Food Research International* 187 (2024), p. 114411. ISSN: 0963-9969. DOI: <https://doi.org/10.1016/j.foodres.2024.114411>. URL: <https://www.sciencedirect.com/science/article/pii/S0963996924004812>.

Appendix

A Regression models on normalized ratings

After normalizing ratings and reformulating the problem as a binary classification task in Section 6.4, we also tried regression models on the adjusted mean ratings, without including price. However, the regression models didn't perform very well, likely because cheap and expensive wines have very different rating variances. Ratings for cheap wines are noisier: they fluctuate a lot, while ratings for expensive wines are more tightly centered around zero. As a result, errors on cheap wines can't be directly compared to those on expensive wines, making the MSE metric that the models optimize not meaningful.

Figure 44 presents the performance results from 5-fold cross-validation. CatBoost outperforms the other models, although its R2 value is close to 0.2, indicating that it only explains 20% of the variance, which is quite low. Table 5 summarizes the test set results, where we observe poor performance in R2 too.

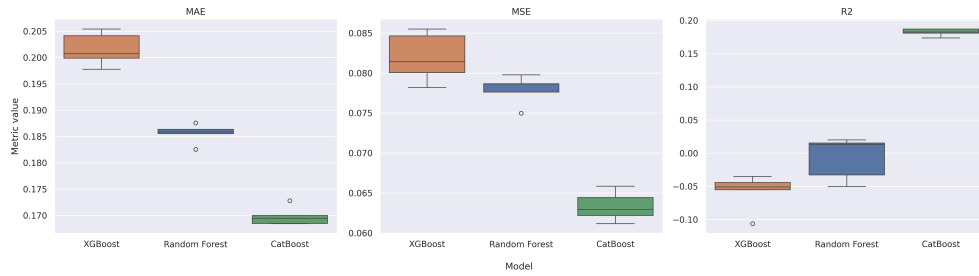


Figure 43: Model performance on adjusted ratings

	MAE	MSE	R2
Train set	0.158	0.0563	0.273
Test set	0.168	0.0612	0.185

Table 5: CatBoost performance on train and test sets

Additionally, Figure 44 illustrates the true versus predicted adjusted ratings, revealing that the model makes accurate predictions for adjusted ratings near 1. However, for the majority of ratings, the model's predictions are clustered between -0.25 and 0.25, even though the true adjusted ratings range from -3 to 2.



Figure 44: True and predicted adjusted ratings